

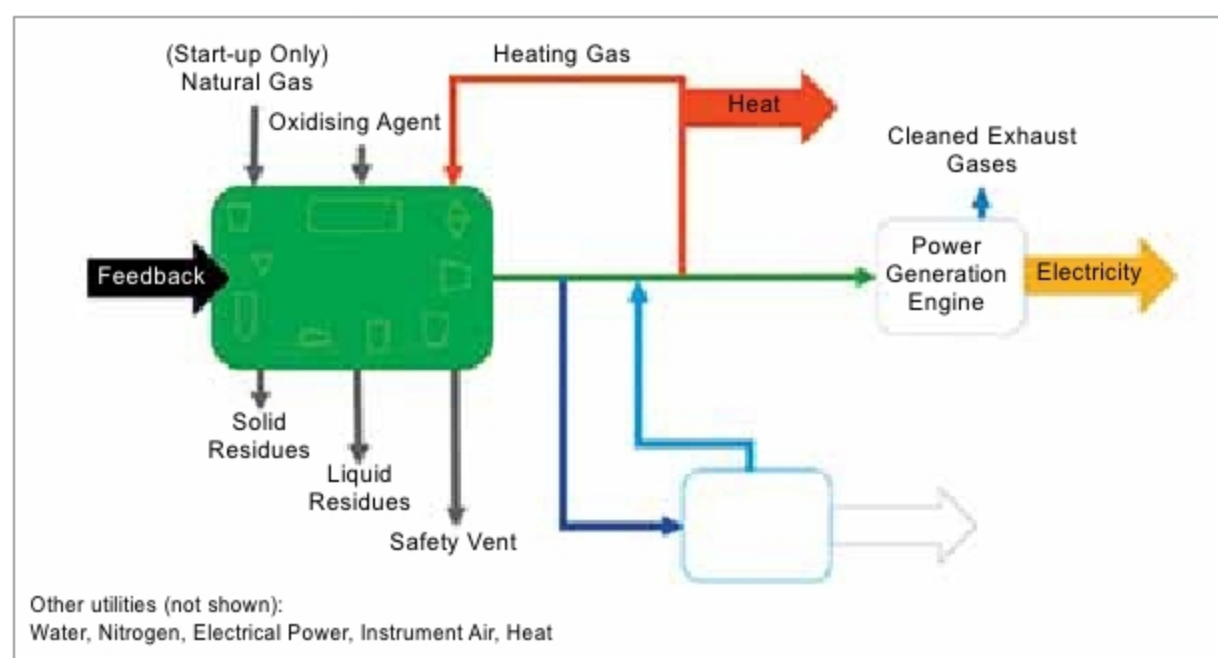


Fig. 3: Waste2Tricity process (Credit: www.waste2tricity.com)

Single-use plastics, also referred to as disposable plastics, are often used in packaging. These are quite harmful and add to the pile of waste more frequently since these are not reused.

But through the process of pyrolysis, a source of power in the form of synthetic fuels can be obtained. For pyrolysis to take place, plastics are heated up in an anaerobic environment (chamber). Renewable energy startups such as India-based Samudhyoga Waste Chakra are incorporating this idea for large scale systems.

In a series of experiments last year, researchers in Singapore were able to successfully convert plastic into formic acid. This chemical acid can be used to generate electricity, which can be used for applications like electric vehicles (EVs).

In another example, researchers from the University of Chester, in collaboration with PowerHouse Energy, have developed a unique method called the Waste2Tricity process that can take the mixed plastic waste and convert it into hydrogen and electricity. The process involves cutting the waste into two-inch strips and melting them together in a kiln. The low carbon hydrogen syngas thus produced can be used as a power source for gas engines while the electricity, which is obtained as a byproduct, can be used to power homes.

There is no need for sorting, and even unrecyclable plastic can be used. Temperatures above 850 degrees Celsius ensure complete gasification and leave low levels of residues. If successful for large scale systems, their

exclusive license will allow them to develop this process in countries such as the UK and Japan.

Betavoltaics

Production of nuclear energy is an extremely powerful yet dangerous process. Nuclear materials emit radioactive particles, which are highly hazardous for organisms as well as the well-being of the environment as a whole. Hence, disposing of nuclear waste to places like landfills, as in the case of other wastes, is not an option.

Scientists have found a way to put these wastes to good use. Betavoltaic devices have the ability to utilise the waste particles produced by low-level radioactive materials for capturing electrons and generating electricity. The electric current is generated from beta particles emitted from a radioactive source. Although the output can last for a long time, its efficiency decreases and hence cannot work on a large scale for now.

Wearables

With the growth of the Internet of Things (IoT), wearables are slowly becoming common around the world. But keeping these wearables charged is not an easy task. Generic batteries are usually too bulky or inflexible to charge sensors for wearables like smart fabrics.

Researchers in China have developed a wearable generator that can convert the movement of human limbs into electricity. The design uses a nickel cantilever and a magnet on a flexible base that can be attached to

an individual's arm or leg joint. When a knee or elbow gets bent during movement, the pressure gets exerted onto the base so that the cantilever springs off and onto the magnet. This ensures that resonance takes place at high frequencies every time, even though the frequency of limb movement for the piezoelectric effect is generally low.

Body heat is a potential source of electricity. Research is ongoing for powering wearables from body sweat too. Although harvesting energy from the human body is not new, the primary challenge for most of such implementations to be wide-ranged is the high initial financing costs required.

Way forward

These are not the only ways of generating electricity in a sustainable way. Every now and then we come across new inventions, but the issue is to scale up such systems practically. In one such example, University College Dublin and the University of Texas at Dallas researchers have been exploring peptide-based nanotubes for their potential use within electronic devices as well as for energy harvesting applications. They have been able to improve the alignment and conductivity of the tubes for better efficiency. New methods need to be devised to make such systems feasible and economically viable.

However, due to Covid-19 pandemic, renewable energy generation in India fell nearly by twenty per cent in June-July, largely impacted by a 41.6 per cent decline in wind-powered electricity output. The electricity generation overall is also at a slower pace for the past few months.

More investment is needed to meet capacity addition targets for clean energy. Stringent measures need to be taken for economic revival and bringing back the growth of this sector. For example, state-owned Coal India's arm Central Coalfields is planning to install an eighty megawatt peak (MWp) ground-mounted solar power plant in the ongoing fiscal year. Apart from this, with the gradual lifting of lockdown for economic activities, the country's energy demand is slowly showing signs of recovery. **EFY**